

LSTAT2130 - Introduction to Bayesian Statistics Project 2026

The file `Weight.txt` contains weight measurements for 128 boys and 152 girls, recorded at birth and subsequently on days 7 and 14; monthly from 1 to 6 months; and every six months from 1 year to 30 months. As an example, the first five subjects have the following data series,

Boy	0d	7d	14d	1m	2m	3m	4m	5m	6m	12m	18m	24m	30m
0	2.68	3.06	3.01	3.74	4.79	5.35	5.96	6.47	6.99	8.83	10.3	11.4	12.4
0	2.97	3.46	3.57	3.93	4.77	5.67	6.07	6.69	7.11	9.23	10.4	11.6	12.5
1	3.53	3.88	4.02	4.71	5.80	6.56	7.23	7.54	8.22	10.30	11.6	12.7	13.8
1	2.54	2.65	3.24	3.77	4.62	5.50	6.06	7.03	7.55	9.51	10.8	12.0	12.8
0	2.91	3.20	3.35	4.17	4.67	5.55	6.17	6.74	6.87	9.09	10.5	11.6	12.7

...

with weight (in kg) reported with 3 significant figures and Boy, a 0-1 binary (with 1 for boys and 0 for girls). For subsequent analysis, the measurement times need to be converted to days using

```
t.obs <- c(0,7,14,30*(1:6),30*seq(12,30,by=6))
```

One commonly used model for the growth of children from birth to 8 years assumes that the weight $y_i(t)$ (in kg) of kid i at time t (in days) is distributed as follows:

$$y_i(t) \sim \mathcal{N}(\mu(t), \sigma^2) \quad \text{where} \quad \mu(t) = \beta_0 + \beta_1 t + \beta_2 \left\{ 1 - \exp\left(-\frac{\beta_3}{\beta_2} t\right) \right\}.$$

with $\beta_k > 0$ for all k . For simplicity, we will first ignore the serial correlation between successive measurements and handle them as if they were independent.

QUESTIONS

- Using graphs, visualize the growth of boys and girls over the 30 months of observation and the evolution of the mean weight over time for the two groups. What does it suggest for the corresponding growth pattern?
- Using a Taylor expansion, obtain a linear approximation to $\mu(t)$ for values of t close to zero.
 - Obtain a linear approximation to $\mu(t)$ for large values of t . [Hint: slant asymptote / asymptote oblique].
 - Using these results, give an interpretation to each β_k parameter.
 - Using these interpretations and the graphs for the mean weight evolution in -1-, obtain rough estimates for the β_k 's for boys. Same question for girls.
 - Using the same graphs, come with a rough visual estimation of the standard deviation σ of the weight at a given age. [Hint: most of the probability mass of a Normal lies within 3 std.dev of the mean].
- Consider the following reparametrization to handle the positivity constraints on the β_k 's,

$$\beta_0 = \exp(\gamma_0) ; \beta_1 = \exp(-\gamma_1) ; \beta_2 = \exp(\gamma_2) ; \beta_3 = \exp(-\gamma_3).$$

Let $\gamma = (\gamma_0, \gamma_1, \gamma_2, \gamma_3)$ and $\tau = 1/\sigma^2$.

- (a) Using a joint improper prior for the model parameters,

$$p(\gamma, \tau) \propto \tau^{-1}$$

obtain (up to a multiplicative constant) the analytical form for the joint posterior density of (γ, τ) .

- (b) Write an R function computing $\log p(\gamma, \tau | \mathcal{D})$ for boys.
 (c) Use the R procedure `nlm` to obtain the Laplace approximation $\mathcal{N}(\hat{\theta}, \Sigma_{\hat{\theta}})$ to $p(\theta | \mathcal{D})$ where $\theta = (\gamma, \tau)$. [Hint: use your rough estimates for the β_k 's and τ obtained in -2d- and -2e- to come with reasonable starting values when calling `nlm`.]
 (d) Using -3a-, identify the conditional posterior distribution of $(\tau | \gamma, \mathcal{D})$.

4. Using the results in -3c- and -3d-:

- (a) Describe, using pseudo-code, a Metropolis-within-Gibbs algorithm to sample the joint posterior of θ , with a Metropolis step for vector γ and a Gibbs step for τ .
 (b) Using R (and without using specialized Bayesian libraries to program your MCMC algorithm),
 i. Implement that pseudo-code to sample 10 000 times the joint posterior of the model parameters for boys. (To enable reproducibility of your results, please type `set.seed(123)` just before starting your algorithm) [In Metropolis steps, target an acceptance rate of 40% for univariate proposals and 20% for multivariate ones. Be clever when choosing your initial conditions].
 ii. Visualize your chains and use one diagnostic tool to assess convergence.
 iii. What is the effective sample size for each parameter chain?
 iv. Based on that sample for (γ, τ) , provide point estimates and 95% credible intervals for the **original** model parameters $\beta_0, \dots, \beta_3$. Given the parameter interpretation, what can your conclude?

5. Still focusing on boys:

- (a) Repeat the analysis in -4b- using JAGS, with a model description based on β_k 's (i.e. without the intermediate γ_k 's).
 (b) Compare your estimation results for the β_k 's with those obtained in Q4(b)iv.
 (c) Using your chains and pure R code, visualize and give summary measures for the predictive distribution of the weight of a boy aged 90 days.

6. Consider now the whole data with boys and girls and extend your model in JAGS to include an individual random intercept and possibly distinct model parameters for boys and girls:

$$(y_i(t) | \beta_{0i}) \sim \mathcal{N}(\mu_i(t), \sigma^2) \quad \text{where} \quad \beta_{0i} \sim \mathcal{N}(\beta_0, \sigma_0^2)$$

with conditional mean

$$\mu_i(t) = \begin{cases} \beta_{0i} + \beta_1 t + \beta_2 \left\{ 1 - \exp\left(-\frac{\beta_3}{\beta_2} t\right) \right\} & \text{for girls,} \\ (\beta_{0i} + \delta_0) + (\beta_1 + \delta_1)t + (\beta_2 + \delta_2) \left\{ 1 - \exp\left(-\frac{\beta_3 + \delta_3}{\beta_2 + \delta_2} t\right) \right\} & \text{for boys.} \end{cases}$$

- (a) What does the addition of a random intercept in the model allow to capture?
 (b) Visualize your chains for the non-random parameters and use one diagnostic tool to assess convergence.
 (c) What do the parameter estimates reveal about differences in growth between boys and girls?
 (d) Visualize the predictive distribution and give summary measures for the weight of a randomly selected boy aged 90 days. Compare your results with those in Question 5c.

INSTRUCTIONS

- By **May 22, 2026 at 1:00 p.m.**, each group of students must submit their results to Madeline VAST by uploading the following 2 documents to the MoodleUCL platform:

1. A single PDF file containing the report detailing the answers to all previous questions (using the same structure and numbering as in the questionnaire). The software code must be in appendix and clearly referenced in the main text.

The file should be named using your family names in a row as in the following example:

Smith-Jones-Brown_Report.pdf

2. A single text file containing the commented software code (R and JAGS only) enabling to reproduce the claimed results. Its subdivision must follow the same structure as the questionnaire.

The file must be named using your family names in a row as in the following example:

Smith-Jones-Brown_Rcode.R

There is no second chance for this report for a later exam session, see below.
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- Each group of 3 students must work independently and without the help of external persons or resources (such as artificial intelligence tools)!! Any fraud detected will result in a severe penalty.
- Any change in group composition not approved by the teachers will result in a zero score for your project.
- Each member of a group must work on all aspects of the project (no “specialization”).

YOUR FINAL MARK FOR THIS COURSE:

Your final mark (*E*: max 20 points) for LSTAT2130 will be obtained by rounding the sum of the results to the nearest integer at:

- the written exam (*W*: max 15 points) in June or August-September ;
- the project (*P*: max 5 points) (written report in May, no second chance for a later session):